WRENFORD BOROUGH COUNCIL · STREETCARE AI

5-Day Team Build Plan · at a glance

5 developers · 48 modules in 10 packages · 6 entry points · tests.py
Build bottom-up: each day only imports what earlier days built

Day 1 · Persistence & data → Day 2 · Knowledge & screening → Day 3 · Features, models & tools → Day 4 · Agents, workflow & chat → Day 5 · Access, API & dashboard

Daily rhythm: 09:00 kick-off · build + unit tests · 13:00 merge · integrate · 16:30 run the entry point

	Day 1 Persistence & raw data cleaning	Day 2 Knowledge base & screening	Day 3 Feature builder, 3 models & tools	Day 4 10-node workflow & chat	Day 5 Access, REST API & dashboard
GOAL	6 tables + 16 accounts; 5 extracts cleaned into **processed/** + quality report	16 PDFs → 136 sections → Chroma; standards read at run time; every report screenable	3 gated models + schemas + metrics; every deterministic tool for the agents	7 single-call agents; graph pauses at the gate and resumes after a restart; guarded chat	Every route guarded; API :8081; dashboard :8080; a review opened and decided end to end
Dev 1 Storage & platform	`persistence/database.py`, `models.py`: engine (multi-thread) + 6 ORM tables, exact column names	`pdf_extraction` + `chunk_splitting`: header line, numbered sections, 700-char passages, 120 overlap, `carried_characters`	`feature_builder`: requirement counts, trigger flags, feature frame, pipeline (median + scale · most_frequent + one-hot)	`state.py` (23 keys, additive call count) + `language_model.py` + **fake LLM & temp DB fixtures** (morning)	`access/permissions` (401 no token · 403 role / district) + `api/schemas` (6 types, 11 models)
Dev 2 Records & retrieval	`repository.py` (13 fns) + `password_security` (PBKDF2, 200k, salt) ✔ EP1 run_database_setup	`vector_index` + `retrieval_tools`: cosine, rebuild replaces, section lookup by number, casebook excluded ✔ EP4 run_knowledge_indexing	`model_registry` + `model_evaluation` (AUC, Brier, R^2, MAE, F1s) + `review_document_tools`	Agents 1–5: plan, requirements, evidence, predictions, recommendation (decision must be open)	`routes.py` A: resources, routers, sign-in/out, queue, detail, 502 handler ✔ EP5 run_api_service
Dev 3 Rules & models	`dataset_loader` + cleaning A: duplicates, refusals, mappings, references → SC+7 digits, whole numbers, latest revision	`standard_reading`: **§2** defect type, **§4** requirements → kind + Decimal limits; refuse wrong type	`repeat_model` (LogReg, AUC ≥ 0.80) + `crew_hours_model` (RF 300, R^2 ≥ 0.86) + behavioural checks ✔ EP3 run_model_training	Agents 6–7: works order / inspection request, resident update (sign-off, no figures, no staff) + `graph.py`	`routes.py` B: reviews (201), decision (lock, 409), calls, chat, audit
Dev 4 Text & live data	Cleaning B: readings (Decimal, half up once), min/max, markup → text, capitals; sentence / document / register cleaners	`report_data_tools` (40 reports, queue Footway-first) + street works count + 3 coordination triggers	`sentence_model` (TF-IDF + LogReg, macro F1 ≥ 0.95) + `evidence_tools` (types, links, disagreement flags)	`nodes.py`: `gather_report_context`, 10 nodes, router, one-transaction writes at nodes 6 & 10	Dashboard A: `api_client`, sign-in, queue, detail ✔ EP6 run_dashboard
Dev 5 Integration & quality	**First hour: rotate keys, .gitignore.** History / repair-log / report cleaners + `quality_profiler` ✔ EP2 run_data_pipeline	`screen_report`: Met / Not met / Not documented, own then linked standard, assessment + decisions open	`prediction_tools` (bands 0.20/0.40, overrun × 1.3) + `decision_tools` (conflicts, refusal order)	`chat/` ×3: memory (3 turns), document answers (≤ 0.64), role-aware record answers, group size ≥ 5	Dashboard B: decision queue, audit trail, chat page + **tests.py** integration & end-to-end
DONE WHEN	EP1: 16 created, rerun 0 / 16. EP2: 32,738→30,569 · 8,202→7,648 · 26,095→25,537 · 6,476→6,337 · 36,661→35,717	16 documents, 136 sections; five different §6 texts; §13 worked example → INSPECT ON SITE; all 40 reports screen	All 3 models above minimums, nothing saved otherwise; live likelihoods in all 3 bands; refusal order tested	Fake-LLM run: 5 calls → pause → new process resumes → 7 / 6 / 5 calls; chat refusals & privacy rules pass	Role × route matrix; open (5 calls) → a second inspector of the same district orders (7 calls); restart mid-review still completes

RULES FOR EVERYONE

- Names exactly as in the specification (a marking suite imports them); every constant from `config.py`.
- No requirement figure in code or config; refuse unknown values with the configured message, never guess.
- Return new tables; exact `Decimal`, round half up once; plain values in API replies; tests with every part.

WHY THIS ORDER

- Training (EP3) calls `screen_report` → needs standards + screening from Day 2, so EP4 runs before EP3.
- Day 4 nodes load the Day 3 models on import; Day 5 API needs every required file.
- Build callers against the spec's signatures; a 5-line stub unblocks parallel work.

STOP THE LINE IF

- An entry point's numbers differ from the spec (row counts, 16 / 136, 16 accounts, 5 / 6 / 7 calls).
- A secret is in git, or a model/document figure appears in a works order or resident update.
- Friday: all 6 entry points from a clean checkout + `pytest tests.py` green.